# QUARTERLY EXPLORATION AND MINING ACTIVITIES REPORT

## July to September 2022 (Q1 FY23)

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**26 October 2022**

Mineral Resources Limited (**ASX: MIN**) (**MinRes** or the **Company**) is pleased to present its Quarterly Exploration and Mining Activities Report for the period to 30 September 2022.

# Q1 KEY POINTS

- Safety performance for the quarter remained strong with no Lost Time Injuries and a Total Reportable Injury Frequency Rate of 2.43, a marginal increase quarter on quarter (qoq).
- Mining Services production volumes were 69Mt during the quarter, 8% higher qoq.
- Iron ore shipments were 4.5M wet metric tonnes (wmt) during the quarter, down 3% qoq, in line with the mine plan and FY23 guidance (17.2 – 18.8M wmt).
- Average realised iron ore price for the quarter was US$72.77 per dry metric tonne (dmt), 15% lower qoq and representing a 70% realisation to the Platts 62% IODEX.
- MinRes announced[1] a Final Investment Decision (FID) to develop the Red Hill Iron Ore Joint Venture assets as part of the Company's Onslow Iron Project formally known as the Ashburton Hub Development.
- Mt Marion and Wodgina remain on track to achieve FY23 production guidance (490 – 540 kdmt attributable[2]).
- The average spodumene concentrate 6% (SC6) equivalent FOB reference price to be applied to Wodgina and Mt Marion shipments of spodumene to be converted to higher value products is US$4,187/dmt for shipments from 1 July 2022 to 31 December 2022. The formula used to calculate this price for each financial half year is 95% of the average monthly price for the 3-month trailing set of three indices (Fastmarkets, Asian Metals and Benchmark Minerals) and will be rebased every six months. This formula is consistent with the publicly stated formula adopted under the Albemarle Offtake Agreement for the Greenbushes mine[3] and used to determine tax and royalty payments for the Australian Tax Office and the Department of Mines Industry Regulation and Safety.
- Mt Marion shipped 56k dmt (51% share) of spodumene concentrate over the quarter. The average realised spodumene concentrate price at Mt Marion was US$2,364/dmt which includes grade adjustments and product discounts.
- The Wodgina ramp up continued with first spodumene concentrate produced from Train 2 in July. Wodgina shipped 26k dmt (40% share) of SC6 over the quarter. Wodgina spodumene concentrate earnings will be recognised once converted to lithium hydroxide and sold.
- A total of 4,703 tonnes (attributable) of lithium hydroxide was converted during the quarter.
    - MinRes' 51% offtake share of Mt Marion spodumene concentrate for the quarter was converted into 3,772 tonnes of lithium hydroxide in China and sold under the tolling agreement with Jiangxi Ganfeng Lithium Co. Ltd (Ganfeng), at an average realised lithium hydroxide price of US$79,288/t (inclusive of China VAT).
    - A total of 931 tonnes of lithium hydroxide was converted by Albemarle from Wodgina spodumene (MinRes 40% share) during the quarter which is expected to be sold in Q2 FY23.
- MinRes continued[4] to evaluate various strategic options to maximise value creation for shareholders, including in relation to its lithium business.
- MinRes expects to commence the next phase of drilling in the onshore Perth Basin in December 2022 following regulatory approval for the North Erregulla 1, Lockyer 2 and Lockyer 3 wells.

[1] ASX Announcement on 29 August 2022
[2] Subject to finalisation of MARBL JV agreement as detailed in the FY22 Full Year Results Presentation dated 29 August 2022
[3] Tianqi Lithium Corporation Global Offering Prospectus (Competent Persons Report), dated 30 June 2022
[4] ASX Announcement on 9 September 2022

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# IRON ORE

| | Share | Units | Q1 FY23 | *QoQ Var* | *PcP Var* |
|---|---|---|---|---|---|
| **Yilgarn Hub** | | | | | |
| Mined | 100% | k wmt | 1,929 | *(4%)* | *(3%)* |
| Produced | 100% | k wmt | 1,972 | *(5%)* | *(24%)* |
| Shipped | 100% | k wmt | 1,657 | *(22%)* | *(30%)* |
| **Utah Point Hub** | | | | | |
| Mined | 100% | k wmt | 2,582 | *(19%)* | *(9%)* |
| Produced | 100% | k wmt | 2,907 | *(2%)* | *19%* |
| Shipped | 100% | k wmt | 2,885 | *13%* | *12%* |

During the quarter, iron ore production and shipments across both hubs were in line with the mine plan and FY23 guidance (17.2 – 18.8M wmt). The Yilgarn processing plant was converted to process both lump and fines, with lump shipments to commence in Q2 FY23.

The average realised iron ore price for the quarter was US$72.77 per dry metric tonne (dmt), 15% lower qoq and representing a 70% realisation of the Platts 62% IODEX. MinRes has commenced the transition of its iron ore sales portfolio from a final quotation period of two months after shipment, to a period of one month after shipment.

# LITHIUM

## Mt Marion

| | Share | Units | Q1 FY23 | *QoQ Var* | *PcP Var* |
|---|---|---|---|---|---|
| **Mt Marion[5]** | | | | | |
| **Spodumene concentrate** | | | | | |
| Mined | 100% | k dmt | 441 | *8%* | *(11%)* |
| Produced | 100% | k dmt | 108 | *(15%)* | *7%* |
| Shipped | 100% | k dmt | 111 | *(22%)* | *56%* |
| Shipped | 51% | k dmt | 56 | *(22%)* | *56%* |
| **Hydroxide** | | | | | |
| Sold | 51% | t | 3,772 | *(44%)* | *N/A* |

Mt Marion production was lower qoq due to continued mining of lower-grade transitional ore and plant shutdowns associated with the expansion of production capacity to 900ktpa (100% basis).

Mt Marion remains on track to achieve FY23 guidance for spodumene concentrate shipped (300-330kdmt at MinRes' 51% share).

MinRes' 51% offtake share of Mt Marion spodumene concentrate for the quarter was toll-converted into 3,772 tonnes of lithium hydroxide and sold under the co-operation agreement with Ganfeng. The average achieved price on the sale of lithium hydroxide for the quarter was US$79,288/t (inclusive of China VAT).

[5] MinRes operates 100% of the Mt Marion project, in which it has a 50% equity interest and a 51% offtake share of spodumene concentrate produced that is currently being toll-converted into lithium hydroxide.

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# Wodgina

| | Share | Units | Q1 FY23 | *QoQ Var* | *PcP Var* |
|---|---|---|---|---|---|
| **Wodgina[6]** | | | | | |
| **Spodumene concentrate** | | | | | |
| Mined | 100% | k dmt | 546 | *795%* | *N/A* |
| Produced | 100% | k dmt | 64 | *221%* | *N/A* |
| Shipped | 100% | k dmt | 66 | *199%* | *N/A* |
| Shipped | 40% | k dmt | 26 | *199%* | *N/A* |
| **Hydroxide** | | | | | |
| Produced | 40% | t | 931 | *N/A* | *N/A* |
| Sold | 40% | t | - | - | - |

Mining operations continued to ramp up, with first spodumene concentrate product for Train 2 achieved in July. During the quarter, practical completion was met on Train 3 and is mechanically available, providing additional flexibility to maximise production and optimise recoveries.

Wodgina remains on track to meet FY23 shipped guidance (190-210kdmt at MinRes 50% share). Wodgina exported three shipments of spodumene concentrate in the quarter, which is expected to be toll-converted into lithium hydroxide in China. Spodumene concentrate earnings will be recognised once converted into lithium hydroxide and sold.

During the quarter, 7k dmt of spodumene concentrate was converted offshore by Albemarle into lithium hydroxide with 931 tonnes of lithium hydroxide produced which is expected to be sold during Q2 FY23.

# Kemerton

Kemerton Train 1 conversion plant is commissioning and expected to produce qualification samples by year end. Kemerton Train 2 conversion plant has achieved mechanical completion and transitioned to commissioning.

[6] MinRes has a 40% equity interest in the project. MinRes continues to work with Albemarle to negotiate binding agreements for the expansion of the MARBL JV. The expansion proposal has been updated since the original announcement of the non-binding agreement with Albemarle. The proposed amendments to the MARBL JV are subject to MinRes and Albemarle negotiating and entering into binding agreements to give effect to the transaction. There is no guarantee that the final agreement will reflect the terms outlined in the FY22 presentation dated 29 August 2022.

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# EXPLORATION AND DEVELOPMENT ACTIVITY

## IRON ORE

### Onslow Iron Project

MinRes announced[7] the Final Investment Decision (FID) to develop the Red Hill Iron Ore Joint Venture assets with joint venture partners Baosteel Resources Australia, POSCO and AMCI. The Onslow Iron Project, previously referred to as the Ashburton Hub Development, will comprise a new mine, processing plant, airport, accommodation resorts, sealed 150km haul road, port, marine infrastructure and transhipping vessel fleet. The Onslow Iron Project is designed to be a 30+ year project at a 30+ Mtpa mining rate with infrastructure capable of 35 Mtpa.

Long lead-time capital equipment has been ordered for the mine, port and marine operations. Mine development activities accelerated throughout the quarter, with construction commencing on the A320-capacity airport, truck maintenance facility, camp and port. Work is progressing on the construction of the transhippers, which are being built at a COSCO yard in China. Four tugs have now been purchased with the fifth agreed in principle pending contract.

Infill drilling continues to progress well at Ken's Bore and Upper Cane deposits with a total 300 reverse circulation (RC) holes for 14,868m and 29 diamond core holes for 1,348m drilled across the deposits.

[7] ASX Announcement on 29 August 2022

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## Yilgarn Hub

The Koolyanobbing Magnetite Phase 1 drilling program continued throughout the quarter. A total of 17,315m of RC and 1,103m of diamond core was drilled. The Phase 1 30,000m program is following up on the promising results that were received from the proof-of-concept drilling program undertaken in FY22. The aim of the Phase 1 program is to scope the resource potential of an initial mining area footprint.

## Pilbara

Project development activities at Marillana deposit continued throughout the quarter with the commencement of a hydrogeology drilling program. Pilot plant geometallurgical test work on the 622t of ore collected through a recent bulk sampling drilling program is ongoing.

# LITHIUM

## Mt Marion

Drilling for the purpose of ore body knowledge and exploration continued throughout the quarter. A total of 692m of diamond core was drilled for metallurgical test work studies to optimise product yield through the processing plant. 6,585m of near mine exploration RC drilling was also completed.

## Wodgina

A drilling program to support the mine restart concluded during the quarter. A total of 5,166m of diamond core and 2,025m of RC was completed. Geo-metallurgical studies to support plant optimisation are ongoing.

# ENERGY

## Perth Basin

During the quarter, the Discovery Assessment Report for Lockyer Deep-1 was prepared and lodged with the Department of Mines, Industry Regulation and Safety (DMIRS), following the successful well test program conducted earlier in the year.

Environmental Plans for the North Erregulla 1 (NED1), Lockyer 2 and Lockyer 3 wells were submitted during the quarter, with approval from DMIRS being granted subsequent to the quarter. The drilling campaign for six wells is planned to commence in December 2022, with either Lockyer 2 or NED1 scheduled first. Drilling remains subject to finalising the rig agreement and related regulatory approvals. The Lockyer 3 and Lockyer 4 wells will follow later in CY23, in addition to two exploration wells elsewhere across MinRes acreage in the Perth Basin. Each well will take approximately 60 days for drilling and mobilisation.

Interpretation of the Romanesque 3D seismic survey dataset acquired over permits EP440, EP389 and Production Licences L18 and L19 continued during the quarter, with several viable drilling targets emerging that if successful, will support the restart of the Red Gully 10 TJ/day gas facility.

## Carnarvon Basin

The joint venture registration process was completed with the regulator for Onslow permit EP510, with MinRes subsidiary Energy Resources Limited installed as operator with 75% ownership, in partnership with Buru Energy Limited (ASX: BRU) (25%). Two wells are planned on this acreage in FY24. A carbon capture and storage feasibility program has also commenced.

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# CONFERENCE CALL

MinRes' investor relations team will be conducting a quarterly results conference call on **26 October 2022** at **8:00am AWST**.

To register use this link: https://s1.c-conf.com/diamondpass/10025848-9hogi8.html. Dial-in details and individual PIN will be received upon registration.

***ENDS***

*This announcement dated 26 October 2022 has been authorised for release to the ASX by Mark Wilson, Company Secretary of Mineral Resources Limited.*

## Further Information

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**About Mineral Resources**
Mineral Resources Limited (ASX: MIN) is a Perth-based leading mining services provider, with a particular focus on the iron ore and lithium sectors in Western Australia. Using technical know-how and an innovative approach to deliver exceptional outcomes, Mineral Resources has become one of the ASX's best-performing contractors since listing in 2006.

To learn more, please visit www.mineralresources.com.au.

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# OPERATING PERFORMANCE SUMMARY

| | Share | Units | Q1 FY23 | Q4 FY22 | Q1 FY22 |
|---|---|---|---|---|---|
| **Safety** | | | | | |
| TRIFR | | | 2.43 | 2.33 | 2.31 |
| LTIFR | | | 0.00 | 0.00 | 0.11 |
| | | | | | |
| **Mining Services** | | | | | |
| Contract Tonnes | | M wmt | 69 | 64 | 75 |
| | | | | | |
| **Yilgarn Hub** | | | | | |
| Iron ore mined | 100% | k wmt | 1,929 | 2,003 | 1,993 |
| Iron ore produced | 100% | k wmt | 1,972 | 2,082 | 2,598 |
| Iron ore sales | 100% | k wmt | 1,657 | 2,129 | 2,371 |
| Fe grade | | % | 56.5% | 56.9% | 56.7% |
| Realisation | | % | 69% | 60% | 46% |
| Revenue | | US$/dmt | 71.0 | 83.4 | 74.8 |
| Moisture | | % | 6.4% | 6.1% | 5.4% |
| Revenue | | $/wmt | 97.5 | 111.1 | 96.3 |
| | | | | | |
| **Utah Point Hub** | | | | | |
| Iron ore mined | 100% | k wmt | 2,582 | 3,196 | 2,835 |
| Iron ore produced | 100% | k wmt | 2,907 | 2,953 | 2,434 |
| Iron ore sales | 100% | k wmt | 2,885 | 2,560 | 2,585 |
| Fe grade | | % | 58.2% | 57.9% | 58.0% |
| Realisation | | % | 71% | 64% | 50% |
| Revenue | | US$/dmt | 73.9 | 88.2 | 81.7 |
| Moisture | | % | 12.8% | 10.9% | 10.3% |
| Revenue | | $/wmt | 92.7 | 110.1 | 99.8 |
| | | | | | |
| **Total Iron Ore** | | | | | |
| Iron ore sales | 100% | k wmt | 4,542 | 4,689 | 4,956 |
| Fe grade | | % | 57.6% | 57.5% | 57.4% |
| Realisation | | % | 70% | 62% | 48% |
| Revenue | | US$/dmt | 72.8 | 85.9 | 78.3 |
| Moisture | | % | 10.5% | 8.7% | 7.9% |
| Revenue | | $/wmt | 94.1 | 122.5 | 95.8 |

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| | Share | Units | Q1 FY23 | Q4 FY22 | Q1 FY22 |
|---|---|---|---|---|---|
| **Mt Marion[8]** | | | | | |
| Spodumene ore mined | 100% | k dmt | 441 | 409 | 496 |
| | | | | | |
| **Spodumene concentrate production:** | | | | | |
| Spodumene concentrate produced | 100% | k dmt | 108 | 128 | 101 |
| Spodumene concentrate shipped | 51% | k dmt | 56 | 72 | 36 |
| High grade product contribution | | % | 24.9% | 7.1% | 47.9% |
| Spodumene concentrate revenue | | US$/dmt | 2,364 | 2,645 | 741 |
| | | | | | |
| **Lithium hydroxide uplift:** | | | | | |
| Lithium hydroxide sold | 51% | t | 3,772 | 6,722 | - |
| Lithium hydroxide revenue[9] | | US$/t | 79,288 | 77,052 | - |
| | | | | | |
| **Wodgina[10]** | | | | | |
| Spodumene ore mined | 100% | k dmt | 546 | 61 | - |
| | | | | | |
| **Spodumene concentrate production:** | | | | | |
| Spodumene concentrate produced | 100% | k dmt | 64 | 20 | - |
| Spodumene concentrate shipped | 40% | k dmt | 26 | 9 | - |
| Spodumene concentrate revenue | | US$/dmt | - | - | - |
| | | | | | |
| **Lithium hydroxide production:** | | | | | |
| Lithium hydroxide produced | 40% | t | 931 | - | - |
| Lithium hydroxide sold | 40% | t | - | - | - |
| Lithium hydroxide revenue | | US$/t | - | - | - |

[8] MinRes operates 100% of the Mt Marion project, in which it has a 50% equity interest and a 51% offtake share of spodumene concentrate produced which is toll-converted into lithium hydroxide.

[9] Revenue on sale of Mt Marion lithium hydroxide is recognised in the Financial Statements as the net earnings under the tolling arrangement, being the achieved price less the cost of spodumene feed, China VAT and conversion. The Lithium hydroxide revenue is shown inclusive of China VAT.

[10] MinRes has a 40% equity interest in the project. MinRes continues to work with Albemarle to negotiate binding agreements for the expansion of the MARBL JV. The expansion proposal has been updated since the original announcement of the non-binding agreement with Albemarle. The proposed amendments to the MARBL JV are subject to MinRes and Albemarle negotiating and entering into binding agreements to give effect to the transaction. There is no guarantee that the final agreement will reflect the terms outlined in the FY22 presentation dated 29 August 2022.